

Lesson 2

Equipment: Raspberry Pi Python code editor, provided sheets, coloured pens/pencils

Lesson Summary

- Pupils will be introduced to Python. Pupils will start by completing a “guess quiz” to gauge an initial understanding of Python concepts. Further guidance on this is provided in a separate document.

Learning Intentions

- I can reflect on the patterns in everyday conversation
- I understand that “input” prompts the user to type a response while “print” simply prints a given message to the console and does not ask for input
- I can reflect on the usage of user input in modern search engines

Curriculum Integration

- TCH 2-13a: pupils engage in a discussion about features of everyday conversation and should identify aspects such as questions and responses. Pupils should then be able to pick out these aspects from their own conversations and “translate” these to Python. For example, identifying a question in their conversation and writing this using Python’s input function.
- TCH 2-14a: pupils should be able to explain the purpose and function of print and input statements in Python.
- TCH 2-15a: pupils create their own “conversation” programs and later engage in discussion evaluating their program as a model of human conversation.

Plan

Starter

- Prompt for pupils: can you have a conversation with a computer?

Quiz (10 minutes)

- Pupils complete a “Python Guess Quiz” in teams of 3-5 people. Further guidance on this is provided in the LanguageTransferGuidance document.

Unplugged Activity (10 minutes)

- Instruct pupils to get into pairs (or a three in case of odd numbers).
- Pupils should engage in a short conversation in their pairs. This should involve pupils asking their partner, the partner answering the question and the pupil responding to their answer.
- Pupils should both ask and answer at least one question.

Taught Section (15 minutes)

- Prompt pupils to find patterns in their conversations. Ask pupils for a few examples of conversations. Try and write these on the board and ask pupils to spot similarities. Pupils should identify a pattern where one person asks a question, the other then answers the questions and finally the first person responds to the given answer. It is recommended that the following constructs should be identified: questions, responses and statements.
- Pupils should then work through their conversations and colour code their conversations in terms of these constructs. For example, highlight questions in blue, responses in red and statements in green.
- On slide 7 there is an example of using Python print and input commands. Use VPC to walk through this example with pupils and ensure pupils understand the difference between print and input. To do this in a VPC style, try swapping print and input in the given example and asking pupils to predict the output.

Coding Activity (30 minutes)

- Pupils write a short program using the example on slide 7 as a template.
- Allow pupils around 10 minutes to first write their code **on paper**. This can be done individually but encourage pupils to work together and help other.
- Pupils then transfer their code to Python and should spend time working on this and noting any errors they had to fix.

Closing (5-10 minutes)

- Pupils engage in a class discussion comparing their programs and experience of “talking” to the computer to human conversation. Pupils should evaluate how close their own conversation program felt to a human conversation.
- [Optional] pupils may benefit from being shown examples of applications which have some form of welcome message and being prompted to consider if this feels similar to interacting with their conversation programs.

Spatial Skills

- Emphasise VPC process in taught section and get pupils to write their code on paper before transferring to the code editor.

General teaching advice

- Pupils may find breaking their conversations down into questions, statements and responses difficult. If pupils find this too challenging, just focus on getting pupils to recognise that input should be used to ask a question.